

Analysis of Algorithms

Quiz-1.

5
6

Q1) for (i = 1; i ≤ n; i++) {
Statement;
}

3.

Time Complexity: $O(n)$

Space complexity: Variable i, n. → 2 var
 $O(1)$ constant.

Q2) for (i = 1; i ≤ n; i = 2) — $\log_2(n)$
{

for (j = 1; j ≤ n; j++) { — n.
print("n")
}

$f(n) = n(\log_2 n)$

Time complexity ~~$O(n)$~~ is $O(n \log n)$

Space complexity: Variable i, j, n ~~$S(n) = 3$~~
 $O(1)$.

Q3) for (i = 1; i ≤ n; i++) { — $n+1$

for (j = 1; j ≤ i; j++) { — $n(n+1)$

for (k = 1; k ≤ n; k = 2) { — $(\log n) \times n$

print("n")
}

$f(n) = (n+1) + n^2 + n + n^2 \log n$

$n^2 + 2n + 1 + n^2 \log n$

Time complexity: $O(n^2 \log n)$ Space complexity: $O(1)$
since all constants.



Information Technology University

Quiz
Analysis of Algorithms (SE311T)
Spring 2026 - BSSE2023/BSCE22
(Section A)

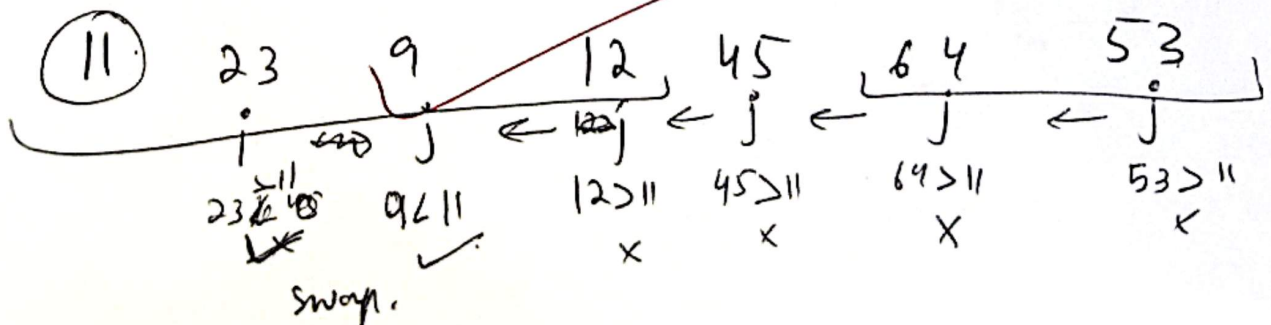
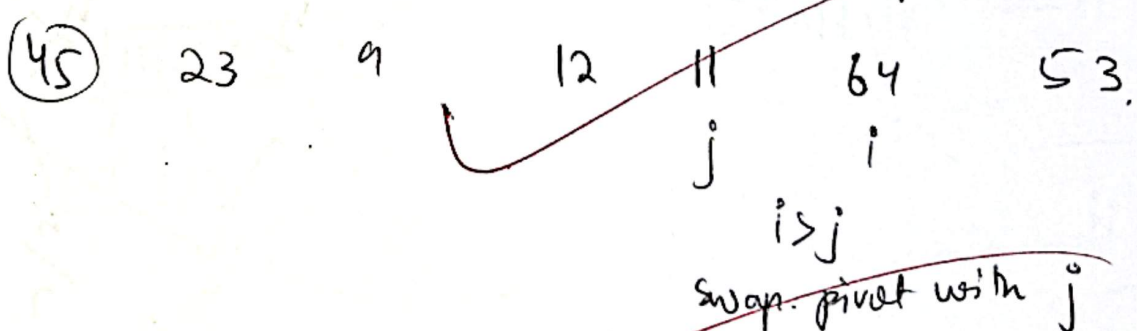
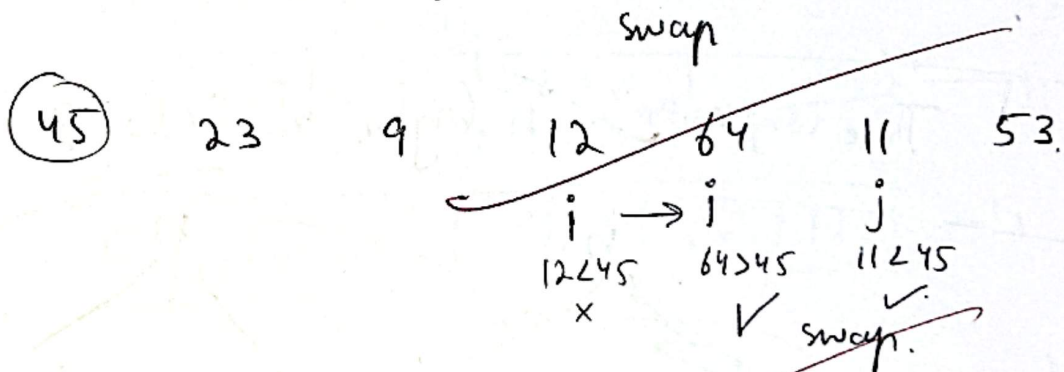
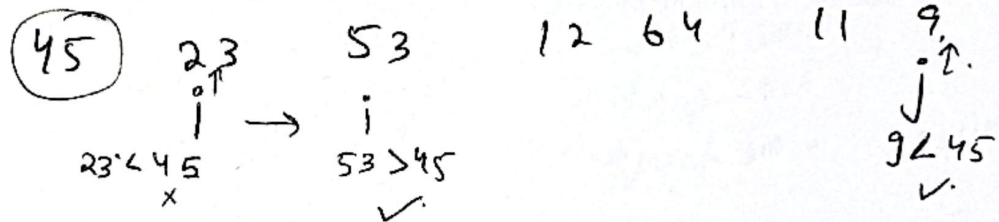
Name Zunaidul Aziz

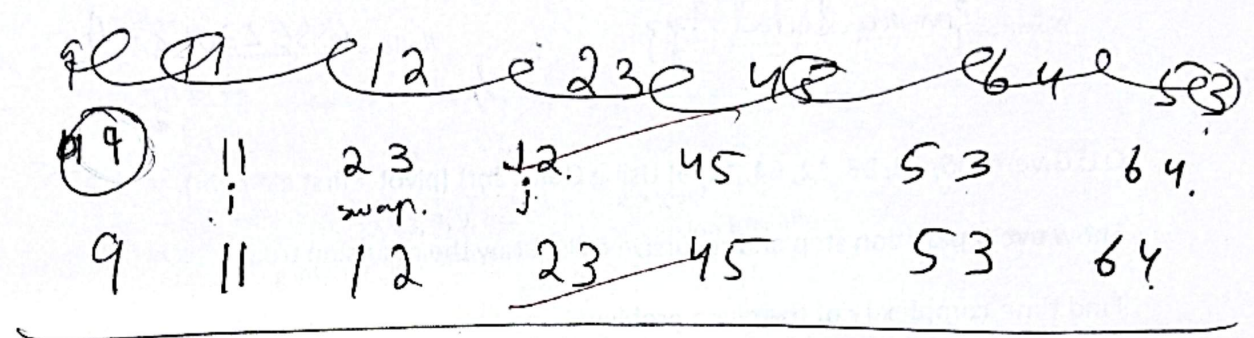
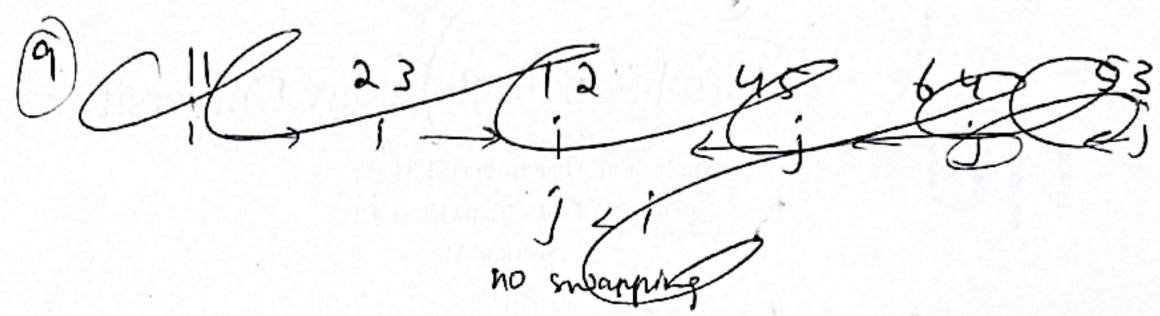
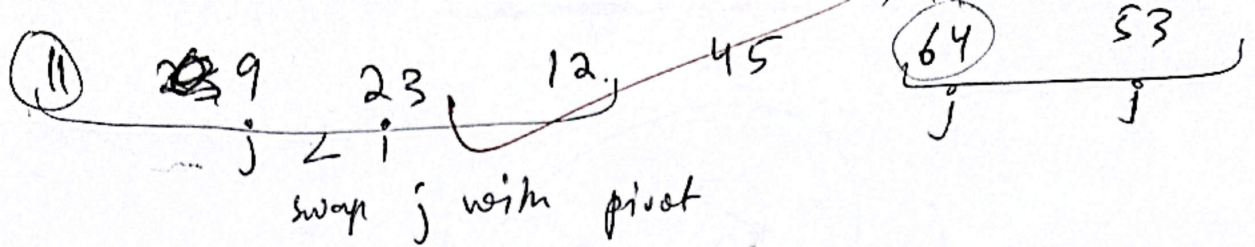
Roll# BSSE23058-A.

Q1: Given: [45, 23, 53, 12, 64, 11, 9] Using Quick Sort (pivot = first element), $j < 45$

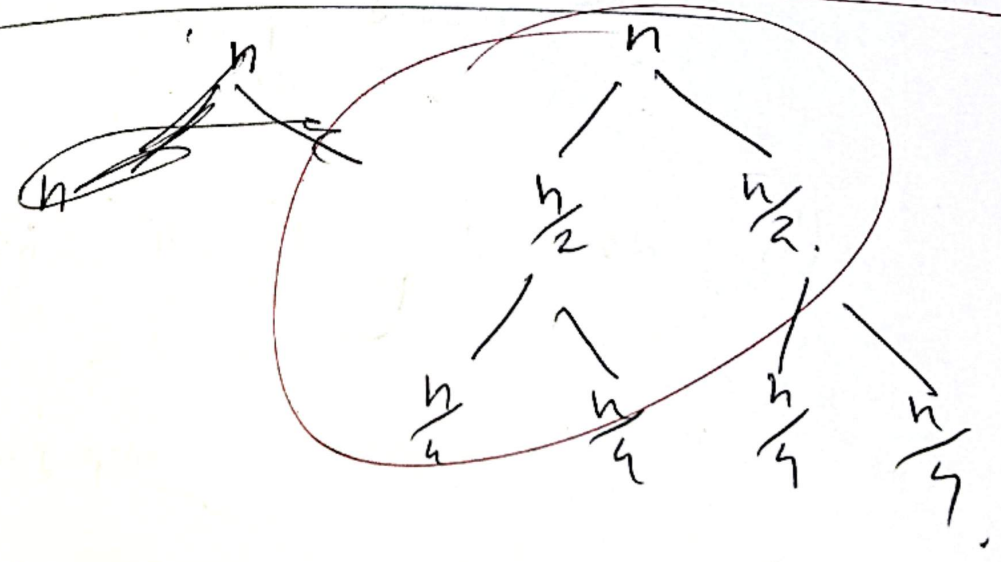
Show every partition step and recursive calls. Draw the recursion tree.

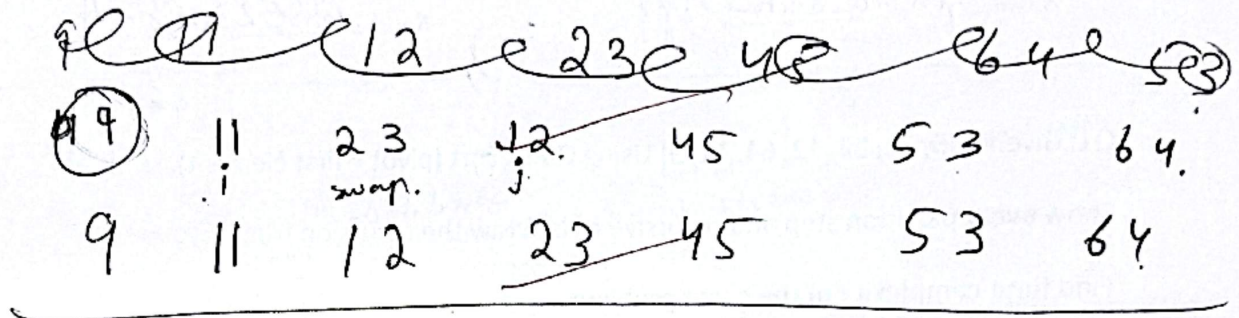
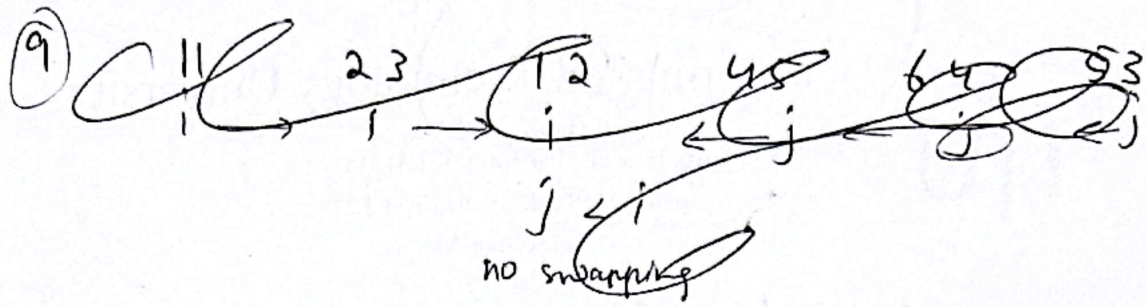
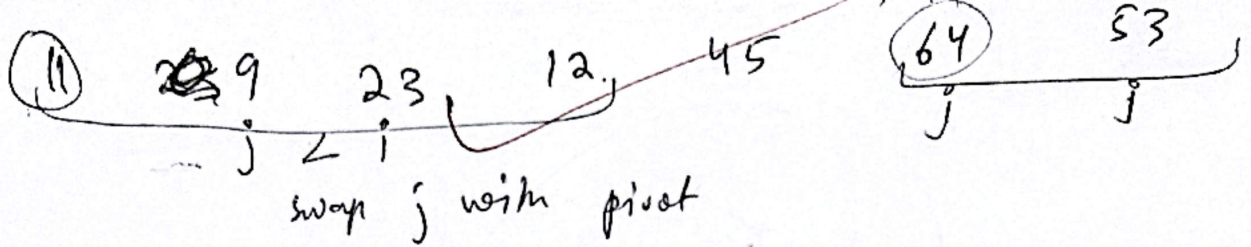
Find time-complexity of the given problem.



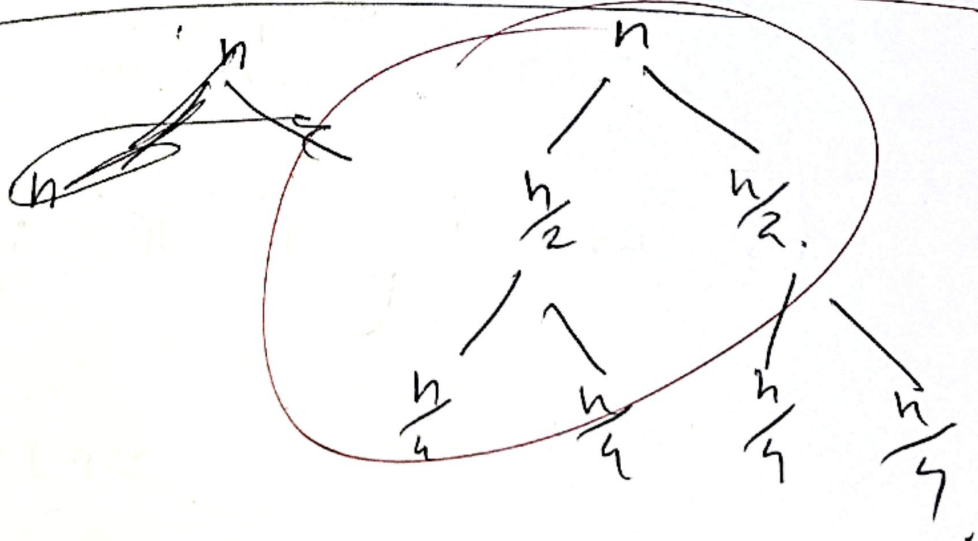


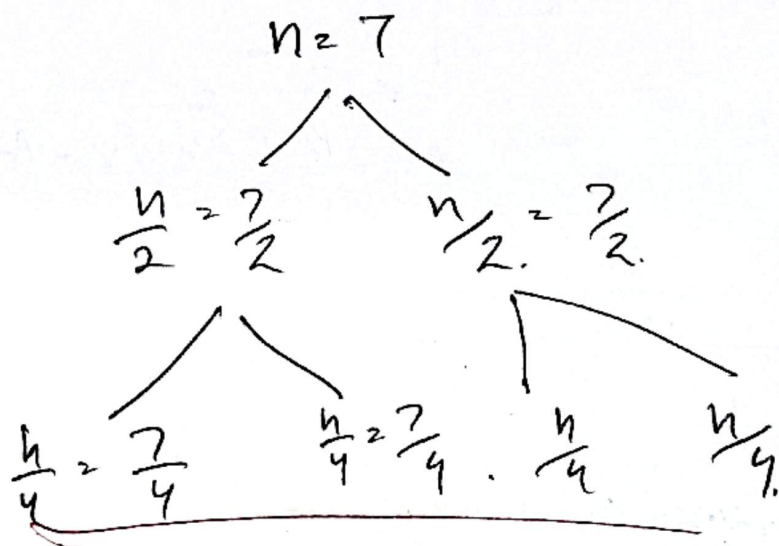
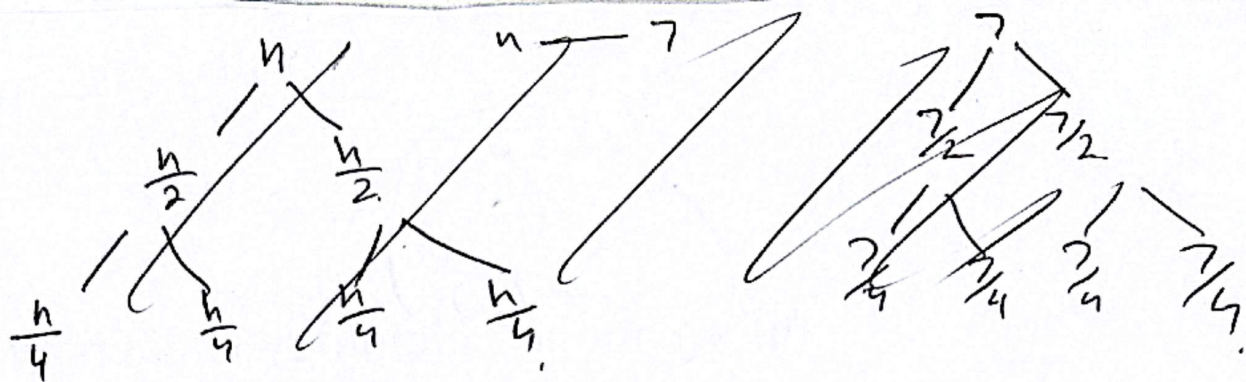
Time complexity $\rightarrow n \log n \approx 7 \log 7 \approx 5.92$





Time complexity $\sim n \log n \approx 7 \log 7 \approx 5.92$





8
10

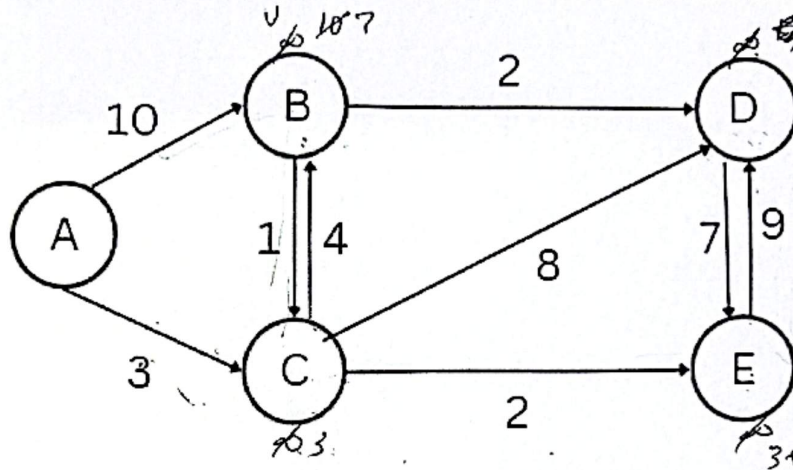
Quiz #4

SE311T/CE416T: Analysis of Algorithms / Algorithm Design and Analysis
Information Technology University of The Punjab, Lahore

Name Zumaira Abdul Aziz
Time allowed: 20 Minutes

Roll# BSS E23058
Maximum Marks: 10

Question #01: Apply single pair shortest path to find shortest path from source vertex a to each of the other vertex of the following directed graph. Also write pseudo code of it and find time complexity of it.



Start from A 10, 3, ∞ , ∞ , ∞

Selected vertex	B	C	D	E	A
B	10	3	∞	∞	∞
C	10	3	12	∞	

3 + 4 = 7
10 > 7
10 > 3
10 + 2 = 12

Selected vertex	B	C	D	E
C	10	3	∞	∞
E	7	3	11	5
B	7	3	9	5
D	7	3	9	5

Time complexity = $O(n^2)$.

Pseudo code:

$$\text{if } (d[u] > d[v] + d[u])$$

- Start at the vertex A.
- Assign the vertex with weights if connected to if not from assign ∞ .
- If $(d[v] > d[v] + d[u])$. If the assigned weight is more than the new calculated weight replace it.
$$d[v] = d[v] + d[u]$$

$u + L(u, v)$
- Select the minimum weighted vertex and repeat the process.

Time

Date: / /Time Complexity: $O(n^2)$

M T W T F S S

$$A^0 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 8 & \infty & 1 \\ \infty & 0 & 1 & \infty \\ 4 & \infty & 0 & \infty \\ \infty & 2 & 9 & 0 \end{bmatrix} \end{matrix}$$

$$\frac{6}{10}$$

$$A^1 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 8 & \infty & 1 \\ \infty & 0 & 1 & \infty \\ 4 & 12 & 0 & 5 \\ \infty & 2 & 9 & 0 \end{bmatrix} \end{matrix}$$

$$A^0[2,3] \quad A^0[2,1] + A^0[1,3]$$

$$1 < \infty + \infty$$

$$A^0[2,4] \quad A^0[2,1] + A^0[1,4]$$

$$\infty = \infty + 1$$

$$A^0[3,4] \quad A^0[3,1] + A^0[1,4]$$

$$\infty > 4 + 1$$

$$\infty > 5$$

$$A^0[3,2] \quad A^0[3,1] + A^0[1,2]$$

$$\infty > 4 + 8$$

$$\infty > 12$$

$$A^0[4,2] \quad A^0[4,1] + A^0[1,2]$$

$$2 < \infty + 8$$

$$A^0[4,3] \quad A^0[4,1] + A^0[1,3]$$

$$9 < \infty + \infty$$

$$A^2 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 8 & 9 & 1 \\ \infty & 0 & 1 & \infty \\ 4 & 12 & 0 & 6 \\ \infty & 2 & 3 & 0 \end{bmatrix} \end{matrix}$$

$$A^1[1,3] \quad A^1[1,2] + A^1[2,3]$$

$$\infty > 8 + 1$$

$$\infty > 9$$

$$A^1[1,4] \quad A^1[1,2] + A^1[2,4]$$

$$1 < 8 + \infty$$

$$A^1[4,1] \quad A^1[4,2] + A^1[2,1]$$

$$\infty > 2 + \infty$$

$$A^1[3,4] \quad A^1[3,2] + A^1[2,4]$$

$$5 < 12 + \infty$$

$$A^1[4,3] \quad A^1[4,2] + A^1[2,3]$$

$$9 > 2 + 1$$

$$9 > 3$$

Date: X

M T W T F S S

$$A_2 = \begin{bmatrix} 0 & 8 & 9 & 1 \\ 5 & 0 & 1 & 6 \\ 4 & 12 & 0 & 5 \\ 7 & 2 & 3 & 0 \end{bmatrix}$$

$$A^2[1,2] = A^2[1,3] + A^2[3,2]$$

$$8 < 9 + 12$$

$$A^2[1,4] = A^2[1,3] + A^2[3,4]$$

$$1 < 9 + 5$$

$$A^2[2,1]$$

$$A^2[2,3] + A^2[3,1]$$

$$A^2[2,4]$$

$$2,3 + 3,4$$

$$\infty$$

$$>$$

$$1 + 4$$

$$\infty$$

$$>$$

$$1 + 5$$

$$\infty$$

$$>$$

$$6$$

$$A^2[4,1]$$

$$4,3 + 3,1$$

$$A^2[4,2]$$

$$4,3 + 3,2$$

$$\infty$$

$$>$$

$$0,3 + 4$$

$$2$$

$$<$$

$$3 + 12$$

$$A_4 = \begin{bmatrix} 0 & 8 & 9 & 1 \\ 5 & 0 & 1 & 6 \\ 4 & 7 & 0 & 5 \\ 7 & 2 & 3 & 0 \end{bmatrix}$$

$$1,2$$

$$1,4$$

$$4,2$$

$$8$$

$$<$$

$$7$$

$$1,3$$

$$1,4$$

$$4,3$$

$$9$$

$$7$$

$$3$$

$$2,1$$

$$2,4$$

$$4,1$$

$$5$$

$$6$$

$$2,3$$

$$1$$

$$2,4$$

$$4,3$$

$$3,2$$

$$3,4$$

$$4,2$$

$$1$$

$$6$$

$$12$$

$$>$$

$$5 + 2$$

$$3,1$$

$$3,4$$

$$4,1$$

Final.

$$\begin{bmatrix} 0 & 8 & 9 & 1 \\ 5 & 0 & 1 & 6 \\ 4 & 7 & 0 & 5 \\ 7 & 2 & 3 & 0 \end{bmatrix}$$